



Finding a length in composite shapes

- 1 A quarter-circular sector is cut out of a circle. The quarter-circle has an area of $32\pi \text{ cm}^2$. What was the area of the original circle? Tick **2** correct answers

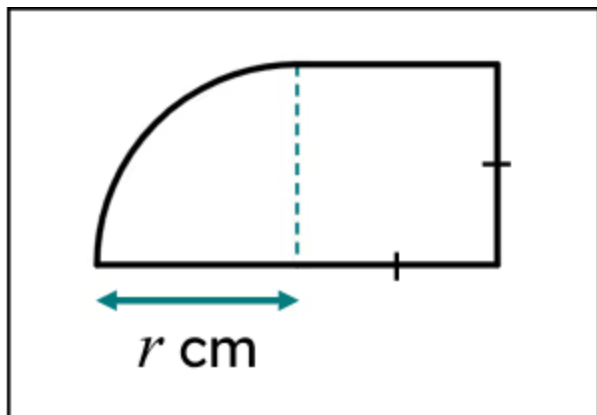
- ☐ $8\pi \text{ cm}^2$
- ☒ $128\pi \text{ cm}^2$
- ☐ $402.12\pi \text{ cm}^2$
- ☐ $16384\pi \text{ cm}^2$
- ☒ 402.12 cm^2

- 2 A semi-circular sector is cut out of a circle. The semicircle has an area of $98\pi \text{ cm}^2$. Which of these statements are true about the original circle? Tick **2** correct answers

- ☐ The original circle has an area of $49\pi \text{ cm}^2$.
- ☒ The original circle has an area of $196\pi \text{ cm}^2$.
- ☐ The original circle has a radius of 7 cm.
- ☒ The original circle has a radius of 14 cm.
- ☐ The original circle has a diameter of 14 cm.

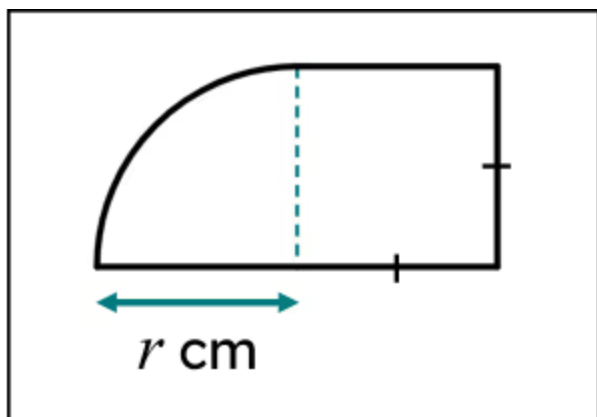


- 3 This composite shape is composed of a quarter-circle and a square. The area of the composite shape is 114 cm^2 . Which of these statements about the shape are correct? Tick 3 correct answers



- ☐ Area of the square is r
- ☒ Area of the square is r^2
- ☒ Area of the quarter-circle is $\frac{1}{4} \times \pi \times r^2$
- Area of the whole shape is
- ☒ $\frac{1}{4} \times \pi \times r^2 + r^2$
- Area of the whole shape is
- ☐ $\frac{1}{4} \times \pi \times r + r$

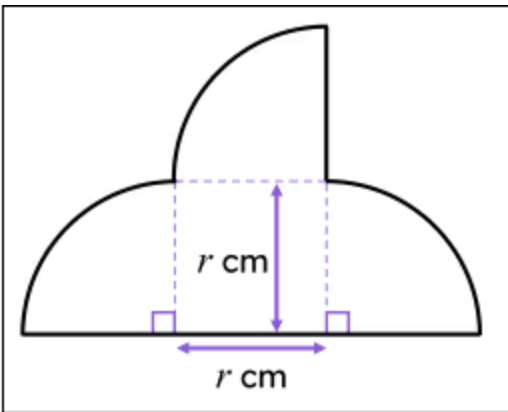
- 4 This composite shape is composed of a quarter-circle and a square. The area of the composite shape is 216 cm^2 . Which of these equations are correct steps in the method to find the radius? Tick 2 correct answers



- ☐ $r \left(\frac{1}{4} \times \pi + 1 \right) = 216$
- ☐ $r \left(\frac{1}{4} \times \pi \times r + r \right) = 216$
- ☒ $r^2 \left(\frac{1}{4} \times \pi + 1 \right) = 216$
- ☐ $r \times \left(\frac{1}{4} \times 4.1416 \dots \right) = 216$
- ☒ $r^2 \times (1.785 \dots) = 216$



- 5 The area of this composite shape is: $(9216\pi + 16384) \text{ cm}^2$. Find the value of r . $r = \underline{128}$.
Fill in the blank



- 6 This composite shape is composed of two quarter-circles and a square. The area of this composite shape is 1360 cm^2 . Find the value of r , rounded to the nearest integer. $r = \underline{23}$
Fill in the blank

